

Project 2 - Cloud Computing

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Git Repository :

<https://github.com/c-demello8/Cloud-Computing-Project/tree/Adding/initial-files>

Abstract

A deployment package was developed for the processing and distribution of the Atlas database data for carrying out analysis on the Higgs Boson experimental data. Constant communication and in-synchronization between the different parts of code was built using Rabbit-MQ and Docker. This report discusses the bottlenecks and the processes of the development of a code suitable for processing large data.

1 Introduction

Understanding the building block of matter and the forces involved in interactions. The standard model was used in Physics for states of matter, in this model all states of matter were massless. In the 1960s Francis Englert and co looked to develop a theory to change this, at the core of this theory lies the Higgs field, symmetry has always been a guiding principle in physics. University of Edinburgh [2026] In 1964 he proposed that a new field fills the entire universe and gives mass to all elementary units. The Higgs boson is a wave in a field, and this proved the concept of the Higgs field. In 2010 the first proton-proton collisions took place at 7 TeV in 2010 with a mass of 125 GeV at atlas and CMS. Nicolaïdou and Sirois [2015] this experiment was the first of its kind and this theory gave strong foundation for particle physics and was proven in 2012 at the Large Hadron Collider. Nicolaïdou and Sirois [2015]

Atlas is one of the two largest detectors at the Large Hadron Collider and is a major focal point. Atlas conducts a range of physics experiments and more specifically the Higgs boson experiment. The Higgs boson and Large Hadron have been the forefront for particle physics. Atlas [2026] Beams of particles from LHC collide with centre of the ATLAS detector and this forms new particles. These particles while then disperse in different directions. and allow for detection of certain properties of these particles, their paths momentums and energies which have led to huge scientific discoveries.

Cloud computing has been innovating the IT infrastructure by reducing upfront cost and financial risk and allows code to be readily deployed without any other requirements. Rage [2025] Docker and RabbitMQ allow for networks to be built and reduce the bottlenecks of normal computing for large experimental data such as the Higgs Boson experiment. This project uses both docker and Rabbit MQ to reduce computational time and produce a working model to analyse the data from the experiments.

2 Method

2.1 Notebook Method

The notebook can be categorized into the following parts for analysis, a priority of this project was to understand the individual components of the code and break them down into their components for reproducibility and this was important for the overall efficiency of the overall project.

Load Data → Clean Process → Perform Calculations
→ Plot Histogram → Interpret Results

The initial notebook provided from the git repository from atlas can be found on the HZZ Analysis Notebook The sections of the code were analysed for the essential parts and stripped and taken for use in our notebook so, in the project the essential parts were only used for the code to reduce the effective run-time of unnecessary code. This can be seen in the notebook.ipynb in the repository this, contains the essential part of code for reproducibility.

The notebook file workbook showcases the plan for the project and the three different sections of the code, the producer/dataloader, the experimentalist/calculator and finally the plotter which takes all the data from the calculator and produces a working histogram of the data. The main section of the code that allows for the parallelization of the code, is the concept that there are jobs present and different URL's which can be analysed separately and collected to produce a histogram.

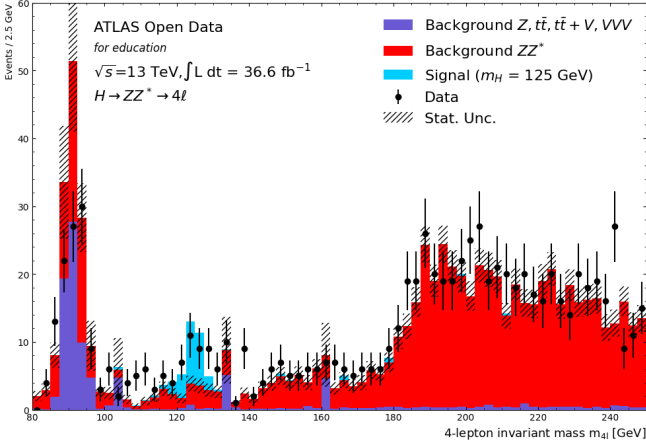


Figure 1: The original histogram of the HZZ analysis.

The histogram produced by the original notebook can be seen in Figure 1. The final output from the containerized program should match this result to show that the program has been executed correctly.

In this project this was a key aim for reproducibility for a deployment package with addition to considering the scalability of the project. Reproduction of the code in this way required a way of connecting the three different parts of the code, a distributor worker and a receiver and Docker and RabbitMQ offered a potential platform for this. RabbitMQ is a powerful and efficient open-source software for message and streaming brokers that enables efficient and reliable communication between applications. RabbitMQ [2026]

2.2 Setting up RabbitMQ and the files

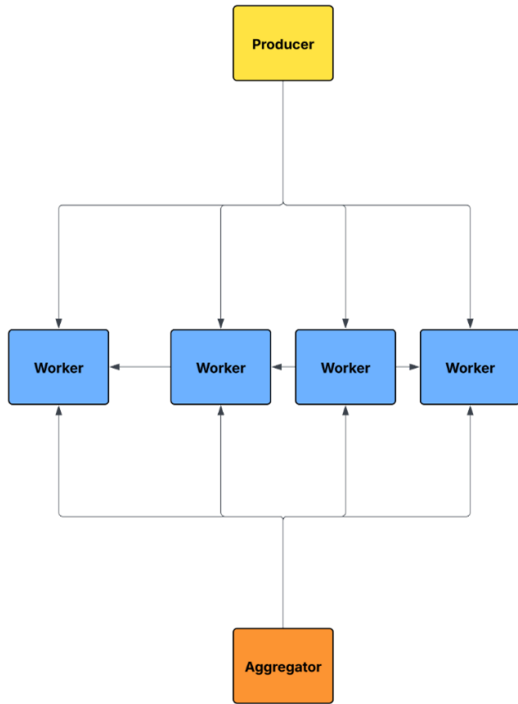


Figure 2: The RabbitMQ Structure, Producer creates jobs and distributes it to each worker they complete the job and give their cleaned data to producers.

The program had to be well structured to perform and reproduce the histograms to utilize docker in combination with RabbitMQ and has to be orchestrated as seen

in Figure 2. Parts of the code that caused problems, are that the awkward arrays and tree data are large datasets. sending large data across the network can lead to crashing. The code had to be altered various times to have settings such as force-restart for the code to carry on running hindering any stopping before the data was fully processed. RabbitMQ allowed for the project to have a flow of data, and structure so that the code could work together in a team to produce the histogram reducing potential slow processing

In the requirements txt you can see the modules that were used in association to make this possible. Important modules that had to be considered for data transfer is JSON. JSON module implements binary protocols for serializing and de-serializing python objects and was essential for transferring data across in the experimental and making it possible for data to be distributed. JSON

JSON data from a reliable source is safe, JSON allowed us to distribute jobs and the processed data for the graphs. JSON The graph from the original data can be reproduced from the working code and in terms of scalability the code can be upscaled by increasing the number of workers in correlation to the number of jobs.

3 Critical Analysis

In terms of the overall use and programming of the code the project successfully containerized a working model of the Higgs boson data and was able to produce a histogram it utilized RabbitMQ's functionality to build a flowing architecture with the use of an aggregator worker and producer. In which the code can be easily accessed and modified by future collaborators to incorporate other docker technologies such as Docker-Swarm and Kubernetes. Kubernetes [2026]

Docker Swarm can be directly implemented in the docker compose file and parameters can be changed so that, replicas of the workers can be used instead of manually inputting the scale of workers. Docker Swarm is an advanced feature for managing a cluster of daemons and would potentially increase the overall automation of the code. Docker [2026] In consideration for real-world application and larger scale production Kubernetes can be performed on the code.

Kubernetes is an application for deployment, scaling and management of containerized applications It has many uses in industry and works closely with Amazons AWS cloud service and also with Microsoft's Azure service. Kubernetes [2026] Kubernetes has a successful roll-back which allows it to carry on running even if one component in the code structure breaks. This is much better for building a fully automated system but requires much more intensive programming.

Papers have showcased that through the use of Kubernetes and AWS achieved a score of 87 percent for its ability in cloud computing compared to 77 percent Alkhatib et al. [2025], for a larger model Kuberens are highly recommended. All in all the project was able to showcase how docker can be used to process large sets of data and deployment packages can be used to make a read to run application for carrying out HZZ analysis.

References

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